Assignment : Data Analytics
(Time : 4-5 hours)

Context

LivingThings installs IoT devices on multiple AC units per site to monitor **real-time AC operation**. Each device records **environmental conditions, AC setpoints, and power consumption**.

Your goal is to perform **deep analytics** to generate operational insights, detect anomalies, identify patterns, and suggest data-driven strategies for optimization and monitoring.

Datasets Provided

1. **sites.csv** → 20 sites (location, type, floor area).
2. **devices.csv** → 60 devices (multiple per site) with model, firmware, **AC tonnage (1, 1.5, 2, 3)**, star rating.
3. **readings.csv** → Hourly readings for March 2024 including `room_temp`, `humidity`, `setpoint`, `outside_temp`, `power_consumption`.

Advanced Analytics Tasks

Part 1: Data Quality & Preprocessing (20–30 mins)

- Detect missing data, duplicates, or outliers.
- Propose **data correction, interpolation, or pipeline improvements**.
- Analyze **sensor drift or inconsistent readings** across devices.

Part 2: Device & Site Consumption Analysis (45–60 mins)

1. **Device-level consumption profiling:**
    - Compute average, median, peak, and variance of power consumption per device.
    - Identify devices with **high variability** and correlate with AC tonnage and star rating.
    - Detect devices that **frequently operate above expected thresholds** based on environmental conditions.
2. **Site-level consumption patterns:**
    - Aggregate per site and identify sites with **abnormally high or irregular consumption**.
    - Correlate consumption with **floor area, site type, number of ACs, and occupancy patterns** (assume uniform occupancy).
3. **Temporal analysis:**
    - Examine hourly, daily, and weekly patterns.
    - Detect **peak load periods** and irregular spikes.

- Apply **time-series decomposition** to separate trend, seasonality, and noise.

Part 3: Anomaly Detection & Root Cause Analysis (45 mins)

- Detect anomalies using **statistical or machine learning approaches** (e.g., z-score, isolation forest, clustering).
- Categorize anomalies:
    1. High consumption despite low temperature difference
    2. Low cooling but high power usage
    3. Sudden spikes/drops in consumption
- Suggest possible causes: device misconfiguration, maintenance needs, or environmental factors.
- Rank anomalies by **impact/severity**.

Part 4: Correlation & Causal Insights (30–45 mins)

- Identify **drivers of consumption** using correlation or regression analysis.
    - AC tonnage, star rating, outside temperature, setpoint differences
- Suggest **operational or deployment strategies** based on insights.

Part 5: Predictive & Prescriptive Analytics (Optional / Bonus)

- Build a **short-term predictive model** for power consumption (per device or per site).
- Suggest **alerts or thresholds** for proactive monitoring (e.g., AC exceeding expected power range).

Deliverables

1. **Notebook / SQL queries / Excel** showing:
    - Device and site-level consumption analysis
    - Temporal and anomaly analysis
    - Correlation and causal insights
    - Optional: predictive modeling
2. **Dashboard mock-up or slide deck (4–5 slides)** summarizing:
    - Key consumption insights and patterns
    - Anomalies and potential root causes
    - Operational recommendations
    - Suggested monitoring and alert framework
3. **1-page strategic note:**
    - Outline **how you would design a device monitoring & analytics framework** for LivingThings.
    - Include **KPIs, dashboards, alerting strategy, and high-level team/process structure**.